

GAME DESIGN DOCUMENT

Genshin Impact - Gacha and Wish System Analysis

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1. System Overview

Genshin Impact's gacha system, referred to in-game as the Wish system, is the main monetisation mechanic built by HoYoverse. Players spend a currency called Primogems to pull on rotating banners for characters and weapons. Rather than relying purely on RNG at a very low drop rate (0.6%), the system layers on a soft pity acceleration curve and a hard pity guarantee on top of the base probability. This turns what could feel like a completely random gamble into something players can plan around, which is a big part of why it works so well for both free-to-play and paying players.

The problem pure RNG creates at low drop rates is frustration and churn. Players who go 100+ pulls without a 5-star feel the game is unfair and disengage. The pity system converts that unpredictable experience into a bounded investment with a known ceiling, which keeps players spending or saving toward a specific goal rather than giving up.

2. Currency Structure

2.1 Currency Chain

There are four main currencies in the wish system, and they follow a fixed conversion chain from real money down to a single wish.

Currency	Source	Used For
Genesis Crystals	Real money purchases only	Convert 1:1 to Primogems; also used for outfits and cosmetics
Primogems	Gameplay (dailies, events, exploration, Abyss) and Genesis Crystal conversion	Convert into Intertwined Fates or Acquaint Fates for wishes
Intertwined Fates	160 Primogems each	Used on limited character banners, weapon banners, Chronicled Wish
Acquaint Fates	160 Primogems each	Standard (permanent) banner only

The key design choice here is keeping Genesis Crystals (the paid currency) separate from Primogems (the earned currency), but convertible at a 1:1 rate. This means the game can show players a unified Primogem balance without distinguishing what was bought and what was earned, which reduces friction and makes spending feel less salient.

2.2 Real Money Pricing (USD, official rates)

Each wish costs 160 Primogems. At standard purchase rates that works out to roughly \$3.20 per pull at the top tier, though the first-time purchase bonus doubles crystals on each tier and resets annually around September, bringing the effective cost down significantly during that window.

Pack	Genesis Crystals	Cost (USD)	Crystals per \$
Handful	60	\$0.99	60.6
Small Bag	300	\$4.99	60.1
Bag	980	\$14.99	65.4
Big Bag	1980	\$29.99	66.0
Huge Bag	3280	\$49.99	65.6

Chest	6480	\$99.99	64.8
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The Blessing of the Welkin Moon (\$4.99/month) gives 300 Genesis Crystals immediately plus 90 Primogems per day for 30 days - 3,000 Primogems total. At around 3.75 wishes per dollar it is roughly 7x more efficient than buying crystal packs directly, and it also builds a daily login habit through the drip-fed reward structure.

3. Banner Types

There are five banner types in Genshin Impact. Each maintains a completely separate pity counter - progress on one banner does not carry over to a different type. This is important to understand before discussing the pity system.

3.1 Character Event Wish (Limited Banner)

The main banner most players focus on. Features one or two limited 5-star characters for a fixed window of around 3 weeks per phase. Pity carries over between Character Event Wish banners of the same type, so if a banner ends before you hit pity, your counter continues on the next character banner.

3.2 Weapon Event Wish (Epitome Invocation)

Features two rate-up 5-star weapons. Uses the Epitomized Path system where players select one target weapon before pulling. Each time a 5-star pull is not the chosen weapon, a Fate Point is awarded. One Fate Point guarantees the chosen weapon on the next 5-star. Hard pity sits at 80 pulls, with soft pity starting around pull 63. Maximum spend to guarantee a specific weapon is 160 pulls (25,600 Primogems).

3.3 Standard Wish (Wanderlust Invocation)

The permanent banner featuring the fixed pool of standard characters and weapons. No featured items, no 50/50 mechanics and no targeting system. Each 5-star is drawn at random from the full permanent pool. Uses Acquaint Fates rather than Intertwined Fates. Generally not a good use of saved resources for players targeting specific characters.

3.4 Beginner's Wish

Only available for the first 20 pulls on a new account. Guarantees Noelle within the first 10 pulls and offers a slight pull discount. Disappears permanently once the 20-pull limit is reached.

3.5 Chronicled Wish

An event banner that lets players pre-select a specific 5-star from a curated list of older limited characters and weapons. Uses a 50% chance on the first 5-star, with a guarantee on the second pull if the first one misses. Only 1 Fate Point is needed to trigger the guarantee (compared to 2 on the weapon banner). Fate Points reset when the banner ends but pity count carries over to the next Chronicled Wish. Not available every version.

4. Pity System

4.1 How Pity Works

Pity is a pull counter that tracks how many wishes have been made since the last 5-star drop. There are two threshold types. Soft pity is where the pull rate starts climbing beyond the base rate, and hard pity is the absolute guarantee where the next pull is 100% a 5-star.

4.2 Probability by Banner

Banner	Base 5-star rate	Soft pity starts	Hard pity	4-star guarantee
Character Event Wish	0.6%	Pull 74	Pull 90	Every 10
Weapon Event Wish	0.7%	Pull 63	Pull 80	Every 10
Standard Wish	0.6%	Pull 74	Pull 90	Every 10
Chronicled Wish	0.6%	Pull 74	Pull 90	Every 10

4.3 Soft Pity Acceleration (Character Banner)

From pull 1 to 73 the base rate stays at 0.6%. Once soft pity kicks in at pull 74, the rate increases by roughly 6% per pull:

- Pull 74: approximately 6.6% (over 10x the base rate)
- Pull 82: over 50% chance of hitting a 5-star
- Pull 90: 100% guaranteed (hard pity)

Because soft pity accelerates the rate so sharply, the average pull count for a 5-star lands around 62 rather than 90. Most players never reach hard pity. The window from pulls 74 to 85 is where the vast majority of 5-stars actually land.

4.4 The 50/50 System

When a 5-star is obtained on a limited Character Event banner there is a 50% chance it will be the featured limited character and a 50% chance it will be a standard pool character. Losing the 50/50 is not a dead end though - the next 5-star on any future Character Event banner is then guaranteed to be the featured character, and this guarantee carries forward indefinitely across banners until used.

Since Version 5.0, the Capturing Radiance mechanic gives players a chance to receive the featured character even when not on a guarantee, which adjusts the effective featured rate to around 55% rather than exactly 50%.

4.5 Worst-Case Cost to Guarantee a Featured Character

Scenario	Pulls	Primogems	USD (approx)
Win 50/50 at average pity (~62)	~62	~9,920	~\$198
Win 50/50 at hard pity	90	14,400	~\$288
Lose 50/50, win next at average pity	~124	~19,840	~\$397
Lose 50/50, hit hard pity twice (worst case)	180	28,800	~\$576

5. RNG Implementation

Genshin Impact uses true independent randomisation per pull. Each wish has no memory of previous results beyond the pity counter incrementing. External factors like pull timing, in-game location, or previous pull history have no influence on outcomes. The community theory of a 'desire sensor' (the idea that the game detects what you want and reduces your odds of getting it) has been consistently disproven by large-scale pull tracking across well over 260 million recorded pulls on platforms like Paimon.moe.

The pity counter is the only persistent state variable between pulls. It is stored server-side and visible to the player through the in-game wish history screen.

6. Free-to-Play Primogem Income

A consistent F2P player can realistically earn around 9,000 to 10,000 Primogems per 6-week version cycle through the following sources:

- Daily commissions: 60 Primogems per day (roughly 2,520 per 42-day version)
- Limited-time events: typically 2,000 to 3,000 Primogems per cycle
- Spiral Abyss: up to 600 Primogems per cycle on a full clear
- New region exploration, quests, and achievements: variable depending on content
- Version update compensation: typically 600 Primogems per update

That works out to roughly 56 to 63 wishes per version - enough for one soft pity pull per cycle on average, or to save for a guaranteed character approximately every 3 version cycles without spending anything.

7. Design Observations

What I found most interesting about this system when researching it is how much of it is psychological rather than purely mathematical.

The pity system reframes spending from gambling to investing. Rather than asking players to bet on a 0.6% chance, it gives them a known ceiling to work toward. This is a deliberate shift from loss framing to investment framing. Players often spend most heavily right before reaching a pity threshold because the reward feels close and certain, which is exactly the behavioural outcome the system is designed to produce.

The 50/50 doubles the engagement loop in a clever way. A single pity cycle resolves as either a win (the player gets what they wanted and stops spending) or a loss (a guarantee is now active, motivating the player to complete the second cycle). This two-stage loop is particularly effective at converting players who were on the fence about spending.

Separate pity pools per banner type prevent players from accidentally building useful pity on the wrong banner. It is presented as a fairness feature (your character pity does not get eaten by weapon pulls) but it also works as a monetisation guardrail that keeps each banner's economy self-contained.

The Welkin subscription ties its value to daily login behaviour. The 90 Primogem daily reward only materialises if the player logs in every day for 30 days. This is a standard retention mechanic dressed up as good value, and it works because the maths genuinely are good - it is just that the maths only work if the player keeps showing up.

8. Glossary

Term	Definition
Primogem	Primary soft currency earned in-game and via Genesis Crystal conversion
Genesis Crystal	Hard currency purchased with real money, converts 1:1 to Primogems
Intertwined Fate	Wish token for limited and weapon banners, costs 160 Primogems each
Acquaint Fate	Wish token for the Standard banner only
Soft Pity	Pull threshold (74 on character banners) where 5-star rates begin accelerating rapidly
Hard Pity	Absolute guarantee threshold (90 on character banners), 100% 5-star on this pull
50/50	50% chance the 5-star is the featured limited character; losing guarantees the next one
Capturing Radiance	Added in v5.0 - gives a chance at the featured character even without a guarantee active
Epitomized Path	Weapon banner targeting system where 1 Fate Point guarantees your chosen weapon
Constellation	Duplicate copies of a character (C1-C6) that power up the character further
Welkin Moon	Monthly \$4.99 subscription giving 300 Genesis Crystals plus 90 daily Primogems for 30 days